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Course: Calculus Beta

Assignment: MyLab Math - Uge 11

Solve the equation by the method of undetermined coefficients.

$$y'' - 10y' = \cos x$$

The general solution to a nonhomogeneous differential equation $ay'' + by' + cy = G(x)$ is the sum of y_c and y_p , where y_c is the general solution to the complementary homogeneous equation $ay'' + by' + cy = 0$ and y_p is any particular solution to the nonhomogeneous equation.

$$y = y_c + y_p$$

First, find y_c by solving the homogeneous differential equation $y'' - 10y' = 0$. The auxiliary equation for $y'' - 10y' = 0$ is shown below.

$$r^2 - 10r = 0$$

This equation has two real and unequal roots.

$$\begin{aligned} r(r - 10) &= 0 \\ r &= 0 \text{ or } r = 10 \end{aligned}$$

The general solution to a second-order homogeneous differential equation whose auxiliary equation has two unique real roots, r_1 and r_2 , is of the form shown below.

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

Substitute the values of r_1 and r_2 into this form to write the solution to the complementary homogeneous equation $y'' - 10y' = 0$.

$$\begin{aligned} y_c &= c_1 e^{0x} + c_2 e^{10x} \\ y_c &= c_1 + c_2 e^{10x} \end{aligned}$$

Now, find y_p , a particular solution to the given nonhomogeneous equation. The given equation has the form $ay'' + by' + cy = G(x)$.

When $G(x)$ is of the form $A \sin kx$ or $A \cos kx$, y_p has the form $B \cos kx + C \sin kx$. To find the coefficients of y_p , substitute y_p and its derivatives into the given nonhomogeneous differential equation and solve for the unknown coefficients B and C .

Note that in the given $G(x)$, k is 1. Find the first and second derivatives of $y_p = B \cos x + C \sin x$.

$$\begin{aligned} y_p' &= -B \sin x + C \cos x \\ y_p'' &= -B \cos x - C \sin x \end{aligned}$$

Substitute the expressions for y_p'' and y_p' into the given nonhomogeneous equation.

$$\begin{aligned} y'' - 10y' &= \cos x \\ -B \cos x - C \sin x - 10(-B \sin x + C \cos x) &= \cos x \end{aligned}$$

Combine like terms.

$$(-B - 10C) \cos x + (10B - C) \sin x = \cos x$$

Compare the left and right sides of the resulting equation. The coefficient of $\cos x$ on the left side must be equal to the coefficient of $\cos x$ on the right side. Likewise, the coefficient of $\sin x$ on the left side must be equal to the coefficient of $\sin x$ on the right side. Note that the right side only has a $\cos x$ term.

$$\begin{aligned} -B - 10C &= 1 \\ 10B - C &= 0 \end{aligned}$$

Solve the system using the addition method. Multiply the second equation by -10 .

$$\begin{aligned} -B - 10C &= 1 \\ -100B + 10C &= 0 \end{aligned} \quad \text{Multiply both sides of the equation } 10B - C = 0 \text{ by } -10.$$

Use the addition method to eliminate the C -term.

$$\begin{aligned} -B - 10C &= 1 \\ -100B + 10C &= 0 \\ \hline -101B &= 1 \end{aligned} \quad \text{Add.}$$

Solve the equation $-101B = 1$ to find B .

$$B = -\frac{1}{101}$$

Use back substitution to find C. Substitute $-\frac{1}{101}$ for B in one of the original equations in the system and solve for C.

$$C = -\frac{10}{101}$$

Substitute these values into the expression $y_p = B \cos x + C \sin x$ and add the result to $y_c = c_1 + c_2 e^{10x}$ to determine the solution to $y'' - 10y' = \cos x$.

$$y = y_c + y_p$$

$$y = c_1 + c_2 e^{10x} - \frac{1}{101} \cos x - \frac{10}{101} \sin x$$